

# HARKEERAT SINGH

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## EXPERIENCE

### [Cionic](#)

San Francisco, CA

#### Software Engineer Intern

May 2024 – Aug 2024

- Built a [real-time](#), WebGL-based front-end interface to collect and parse biofeedback data from bionic clothing, using surface electromyography electrodes with embedded FES (functional electrical stimulation) to augment lower body muscle stimulation.
- Implemented live signal processing algorithms by utilizing Cionic's research portal to render muscle activation graphics. Leveraged automated post-session data analysis through Cionic's REST APIs and custom data pipelines.

### [Kwon Perception Lab](#)

Berkeley, CA

#### Virtual Reality Research Assistant

Sep 2023 – May 2024

- Architected a VR obstacle course in Unity for studying spatial navigation in amblyopic patients, integrating the HTC VIVE Pro Eye headset for real-time gaze tracking and monocular/binocular navigation testing across 100 randomized layouts.
- Deployed eye-tracking calibration using Unity's custom fixation task and dynamic scoring mechanisms based on proximity to obstacles, collecting metrics on pathing, trial duration, gaze behavior, and risk aversion across 30 trials.

### [The Verse](#)

Berkeley, CA

#### Unity/Unreal 3D Developer Intern

May 2023 - Aug 2023

- Constructed 6 VR game modules to integrate real-time object physics and player feedback loops. Reduced interaction latency by 20% through physics engine optimizations and asset compression techniques.
- Programmed level transition systems using Unreal Engine's Blueprint scripting to control event-driven gameplay mechanics. Designed proximity-triggered event systems for multimedia playback, reducing asset load time by 24%.

### [Extended Reality at Berkeley](#)

Berkeley, CA

#### VR Developer

Aug 2022 - Dec 2022

- Configured real-time object detection systems and integrated visual overlays to allow dynamic interaction between virtual objects and physical environments in augmented and extended reality applications.
- Utilized Unity's scripting interface to implement gesture detection algorithms for a [VR first-person PvP game](#) that tracked positional and orientation data from VR controllers with a 91% pattern accuracy.

## EDUCATION

### University of California, Berkeley

B.S. in Computer Science

Aug 2020 - May 2025

#### Relevant Coursework:

Data Structures and Algorithms | Computer Security | Computer Graphics | Database Systems | Computer Visions | Computational Photography | Artificial Intelligence | Machine Learning | Computer Architecture | Data Science

## PROJECTS

### Smoothed-Particle Hydrodynamics 3D Fluid Simulation

- Established GPU compute shaders in GLSL to manage real-time pressure, velocity, and viscosity calculations for 200,000+ particles using [SPH](#), leveraging Tellusim SDK for cross-platform compatibility across Metal and CUDA.
- Leveraged real-time fluid simulation controls, enabling user interaction with fluid particles using mouse-driven invisible spheres for dynamic fluid movement and real-time visual feedback.

### 3D Real Time Muscle Deformation Toolkit

- Deployed a real-time 3D muscle deformation system that processes live EMG data from 16 electrodes, translating muscle activation signals into interactive deformation on a [3D avatar](#) using signal-driven rendering algorithms.
- Enabled an Exponential Moving Average (EMA) filtering algorithm to smooth EMG data and eliminate high-frequency noise, creating a calibrated input stream for real-time deformation and color interpolation based on normalized muscle activation percentages.

## SKILLS

**Languages/Scripting:** Python | Java | JavaScript | C/C++/C# | GLSL | SQL/NoSQL | HTML | CSS | MATLAB

**Graphics/APIs/Libraries:** Unity | Unreal Engine | WebGL | OpenGL | Vulkan | Three.js | Babylon.js | Blender | Autodesk Maya

**Computer Vision & Machine Learning:** OpenCV | TensorFlow | PyTorch | ARKit | ARCore | Vuforia

**Development Tools:** Node.js | React.js | Docker | Jupyter Notebooks | Git

**Data Analysis:** Pandas | NumPy | SciPy | Matplotlib